

Transport Relaxation Mechanisms in Bilayer Graphene: Effects of Pauli Blocking

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Abstract

We present a unified analytical treatment of carrier transport relaxation in bilayer graphene (BLG), including scattering by surface roughness, charged and neutral impurities, acoustic and optical phonons, and substrate polar phonons. The calculation is formulated within the low-energy two-band approximation using Fermi's golden rule and the semiclassical Boltzmann transport equation. Closed-form expressions for the transport relaxation rates are derived and verified by numerical evaluation of the corresponding scattering integrals. Particular attention is given to Pauli blocking, whose influence becomes important under degenerate carrier conditions owing to the approximately parabolic low-energy dispersion and nearly constant density of states of BLG. We find that Pauli blocking modifies the relaxation rates in a mechanism- and energy-dependent manner for both suspended and substrate-supported BLG. For the parameters considered, neutral-impurity scattering provides the dominant contribution over a substantial energy range, while acoustic-phonon and charged-impurity scattering remain important competing mechanisms. Optical-phonon scattering is suppressed below its emission threshold, whereas substrate polar phonons provide an additional relaxation channel in supported BLG. Comparison with monolayer graphene and a conventional two-dimensional electron gas highlights the role of band dispersion, density of states, screening, and chirality in determining the distinct scattering behavior of BLG. These results provide analytical insight into the relative importance of the principal momentum-relaxation mechanisms and the role of Pauli blocking in BLG transport.

1 Introduction

Graphene is a two-dimensional form of carbon in which sp^2 -bonded carbon atoms form a honeycomb lattice [1]. Although its unusual electronic properties had been predicted theoretically, graphene was first isolated experimentally in 2004 by Geim and Novoselov [2]. Subsequent experiments revealed an unconventional integer quantum Hall effect, demonstrating that carriers in monolayer graphene (MLG) behave as massless chiral quasiparticles with a linear Dirac dispersion [3, 4].

Bilayer graphene (BLG), most commonly realized in the Bernal (AB) stacking configuration, has a markedly different low-energy electronic structure. Interlayer coupling transforms the approximately linear dispersion of MLG into an approximately parabolic low-energy spectrum, so that the carriers behave as massive chiral quasiparticles. This regime is described effectively by a two-band Hamiltonian [5] and gives rise to an integer quantum Hall effect distinct from that of MLG [6, 7].

An important feature of BLG is its electrical tunability. Electrostatic gating controls the carrier density, while a potential difference between the two layers can open and tune a band gap [8–10]. Together with its favorable electronic, thermal, mechanical, and optical properties [5, 11–13], this tunability makes BLG attractive for nanoelectronic, optoelectronic, thermoelectric, valleytronic, and high-frequency applications [13–19].

For practical BLG devices, however, the electronic band structure alone does not determine transport. Carrier momentum relaxation is governed by a variety of disorder- and phonon-induced scattering mecha-

nisms. Important disorder mechanisms include neutral-impurity (NI), surface-roughness (SR), and charged-impurity (CI) scattering, while phonon-mediated processes include acoustic-phonon (AP), optical-phonon (OP), and, in substrate-supported devices, surface-polar-phonon (SPP) scattering [20–23]. Substrate support also modifies the transport environment through dielectric screening and introduces additional remote SPP scattering channels [21, 24–26]. In suspended graphene-based systems, flexural out-of-plane (ZA) phonons can provide an additional low-energy scattering channel, with their contribution depending on membrane tension, anharmonic effects, and environmental constraints [27]. The present analysis, however, includes only longitudinal acoustic phonons within the AP contribution and does not explicitly treat ZA scattering.

The occupation of electronic final states provides another important factor in transport scattering. Pauli blocking (PB), which follows from the Pauli exclusion principle, reduces the phase space available for a scattering transition when the final electronic state is occupied. This effect is particularly relevant to inelastic processes, for which the initial and final electronic states generally have different energies and their occupations are governed by Fermi–Dirac statistics [28, 29]. PB is expected to be most pronounced in the degenerate regime and to weaken as increasing temperature broadens the carrier distribution.

The analysis is restricted to momentum-relaxing disorder and electron–phonon scattering. Normal electron–electron collisions are not included because, in a translationally invariant system, they conserve total electronic momentum and therefore do not directly produce momentum relaxation. Such interactions can nevertheless be important for carrier thermalization and hydrodynamic transport, particularly near charge neutrality [30, 31], but these effects lie outside the scope of the present treatment. Likewise, flexural out-of-plane (ZA) phonon scattering is not included; the AP contribution considered here is restricted to longitudinal modes.

Carrier transport in BLG has been studied extensively for charged impurities, short-range disorder, acoustic and optical phonons, and substrate-related scattering mechanisms [21, 23, 32]. More recent work has considered electron–phonon interactions, carrier cooling, and hydrodynamic transport regimes [30, 31, 33, 34]. PB has also been investigated in detail for inelastic transport in MLG [29]. However, the combined effects of electronic structure, momentum-dependent scattering interactions, finite phonon energies, and final-state PB are not commonly treated within a single analytical framework for low-energy BLG transport.

In this work, we develop a unified analytical and numerical treatment of the principal momentum-relaxing scattering mechanisms considered here in low-energy BLG. NI, SR, CI, AP, OP, and SPP scattering are formulated within a common Boltzmann transport framework. The treatment retains the momentum dependence of finite-range scattering potentials and, for inelastic processes, the finite phonon energies and associated emission and absorption constraints. PB of the final electronic states is incorporated explicitly, allowing its influence on the energy-dependent transport relaxation rates to be quantified. The resulting analytical and semi-analytical expressions are validated by direct numerical evaluation of the corresponding scattering integrals.

We further compare selected scattering rates in BLG with those of MLG and a conventional two-dimensional electron gas (2DEG). This comparison separates the effects of the scattering interaction from those arising from the underlying band dispersion, DOS, and chiral wave-function overlap. Finally, the different scattering channels are evaluated over a common carrier-energy range to identify their relative importance and the regimes in which PB and finite-energy phase-space restrictions substantially modify carrier relaxation. The resulting framework provides a systematic basis for assessing how electronic structure, scattering interaction, and final-state occupation collectively govern transport relaxation in BLG.

2 Formalism

To investigate how different scattering mechanisms affect carrier transport in BLG, we use the semiclassical Boltzmann transport equation (BTE). Within the relaxation-time approximation, the BTE connects the nonequilibrium carrier distribution with the transport relaxation time τ . Its inverse, τ^{-1} , gives the corresponding transport scattering rate. Since not every scattering event contributes equally to electrical resistance, the transport rate includes an angular weighting factor that accounts for the amount of momentum lost in a scattering event. In particular, forward scattering produces little momentum relaxation, whereas backscattering is much more effective [32, 35].

Following Refs. [29, 36], the energy-dependent transport relaxation rate is written as

$$\frac{1}{\tau_k} = \frac{A}{(2\pi)^2} \int k' dk' d\theta_{kk'} (1 - \cos \theta_{kk'}) t_{kk'}^{\text{int}} \frac{1 - f(E_{k'})}{1 - f(E_k)}, \quad (1)$$

where A is the area of the BLG sheet, $t_{kk'}^{\text{int}}$ denotes the interaction-induced transition probability for an electron to scatter from the initial state $\mathbf{k}$ to the final state $\mathbf{k}'$, and $\theta_{kk'}$ is the scattering angle between the two momentum states. The factor $1 - \cos \theta_{kk'}$ is the transport factor and accounts for the efficiency with which a scattering event relaxes the carrier momentum: it vanishes for forward scattering and reaches its maximum for backscattering, $\theta_{kk'} = \pi$.

The ratio $\frac{1-f(E_{k'})}{1-f(E_k)}$ is the PB factor and accounts for the occupation of the final electronic state in the scattering process [29]. For strictly elastic scattering, $E_{k'} = E_k$, and hence the PB factor reduces identically to unity. Its nontrivial contribution in the present formulation therefore arises in inelastic phonon-mediated processes. The electronic occupation is described by the Fermi–Dirac distribution, $f(E) = \frac{1}{1+\exp[\beta(E-\mu)]}$, $\beta = \frac{1}{k_B T}$, where μ is the chemical potential, k_B is the Boltzmann constant, and T is the absolute temperature.

Within the low-energy two-band approximation, BLG is described by an approximately parabolic dispersion, $E_k = \frac{\hbar^2 k^2}{2m^*}$, where m^* is the effective electron mass [5, 32]. The corresponding low-energy spinor wavefunction may be written as $\psi_{\mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{2A}} \begin{pmatrix} e^{i\theta_k} \\ e^{-i\theta_k} \end{pmatrix} e^{i\mathbf{k}\cdot\mathbf{r}}$, where $\mathbf{r}$ is the position and θ_k is the polar angle of the electron wave vector $\mathbf{k}$. The winding-two chiral structure of the BLG spinor gives rise to the overlap factor $u_{kk'} = |\langle u_{\mathbf{k}'} | u_{\mathbf{k}} \rangle|^2 = \frac{1+\cos(2\theta_{kk'})}{2}$, which determines the angular dependence of scattering in the low-energy BLG model [5].

For a scattering process from $\mathbf{k}$ to $\mathbf{k}'$, the transferred momentum is $q = |\mathbf{k}' - \mathbf{k}| = \sqrt{k^2 + k'^2 - 2kk' \cos \theta_{kk'}}$. For elastic scattering, $E_{k'} = E_k$, and hence $k' = k$, so that $q = 2k \sin\left(\frac{\theta_{kk'}}{2}\right)$. For inelastic scattering, k and k' are generally different and are related through the appropriate energy-conservation condition.

The transition probability is evaluated using Fermi's golden rule,

$$t_{kk'}^{\text{int}} = \frac{2\pi}{\hbar} |\langle k' | H_{\text{int}} | k \rangle|^2 \delta(E_f - E_i), \quad (2)$$

where E_i and E_f denote the total initial and final energies of the electron–scattering-system configuration, respectively [32]. For elastic impurity scattering, $E_f - E_i = E_{k'} - E_k$, whereas phonon-mediated scattering additionally involves the absorption or emission of a phonon with energy $\hbar\omega_q$.

For the scattering mechanisms considered here, the squared interaction matrix element can be separated into a mechanism-dependent coupling and the BLG chiral overlap factor,

$$|v_{kk'}|^2 = |M(q)|^2 u_{kk'}, \quad (3)$$

where $M(q)$ contains the mechanism-specific coupling.

For phonon-mediated processes, the electron–phonon interaction can be written in the generic second-quantized form $H_{e-ph} = \sum_{\mathbf{k}, \mathbf{q}} v_{\mathbf{k}\mathbf{q}} c_{\mathbf{k}+\mathbf{q}}^\dagger c_{\mathbf{k}} (b_{\mathbf{q}} + b_{-\mathbf{q}}^\dagger)$, where $c_{\mathbf{k}}^\dagger$ and $c_{\mathbf{k}}$ are electron creation and annihilation operators, while $b_{\mathbf{q}}^\dagger$ and $b_{\mathbf{q}}$ are the corresponding phonon operators [21, 29].

For elastic disorder-induced scattering, the transition rate can be written in the general form

$$t_{kk'}^{\text{dis}} = \frac{2\pi}{\hbar} \mathcal{W}_{\text{dis}}(q) u_{kk'} \delta(E_{k'} - E_k), \quad (4)$$

where $\mathcal{W}_{\text{dis}}(q)$ denotes the mechanism-dependent disorder coupling function. It incorporates the strength and spatial correlations of the disorder and, where appropriate, the effects of dielectric screening. For a specific disorder mechanism, $\mathcal{W}_{\text{dis}}(q)$ is determined by the corresponding disorder potential or disorder correlation function [23, 32]. The specific forms of $\mathcal{W}_{\text{dis}}(q)$ for NI, SR, and CI scattering are derived separately in the corresponding sections.

For phonon-mediated scattering, the transition probability obtained from Fermi's golden rule is

$$t_{kk'}^{\text{ph}} = \frac{2\pi}{\hbar} |v_{kq}|^2 [n_{\text{ph}} \delta(E_k + \hbar\omega_q - E_{k'}) + (n_{\text{ph}} + 1) \delta(E_k - \hbar\omega_q - E_{k'})], \quad (5)$$

where n_{ph} is the equilibrium phonon occupation number, $n_{\text{ph}} = \frac{1}{\exp(\beta\hbar\omega_q)-1}$, which follows the Bose–Einstein distribution [29]. The first term in Eq. (5) describes phonon absorption, for which the electron gains energy $\hbar\omega_q$, whereas the second term describes phonon emission, for which the electron loses the same amount of energy. The Dirac delta functions enforce energy conservation for the respective processes. For OPs with approximately fixed energy $\hbar\omega_o$, the emission term is subject to a finite-energy threshold, whereas AP scattering involves much smaller phonon energies and approaches the quasi-elastic limit when $\hbar\omega_{\text{AP}} \ll k_B T$.

The Coulomb interaction associated with charged scattering centers is modified by dielectric screening. Within the random-phase approximation, the static dielectric function can be written as

$$\epsilon_q = 1 + \frac{2\pi e_c^2}{\epsilon q} \Pi(q), \quad (6)$$

where ϵ denotes the effective background dielectric constant of the BLG–substrate environment and $\Pi(q)$ is the static polarization function of BLG [32, 37]. In the long-wavelength Thomas–Fermi approximation, the static polarization function is replaced by its long-wavelength limit, $\Pi(q) \simeq \Pi(0) \simeq D(E_F)$, where $D(E_F)$ is the DOS at the Fermi energy. This approximation is appropriate in the degenerate, long-wavelength regime. The resulting screened dielectric function enters the matrix elements of scattering mechanisms whose interaction potentials are affected by the dielectric response of the BLG environment, most notably CI scattering and, where applicable, SR scattering.

The scattering mechanisms considered in this work are intended to capture the principal disorder- and phonon-induced processes responsible for carrier momentum relaxation over the energy and temperature ranges studied. The disorder-related mechanisms are NI, SR, and CI scattering. These represent, respectively, short-range neutral disorder, potential fluctuations associated with the interface, and long-range Coulomb disorder. The phonon-related mechanisms are AP, OP, and SPP scatterings, representing intrinsic lattice vibrations and polar phonon modes associated with the substrate. Together, these mechanisms constitute the main elastic and inelastic scattering channels included in the present BLG transport model. Other possible mechanisms are not considered because their treatment would require additional material-specific parameters or would extend beyond the scope of the present analysis.

The corresponding squared interaction coupling functions are introduced below:

$$|v_{kk'}|^2 = \begin{cases} (2\pi a^2 V_0)^2 e^{-q^2 a^2} u_{kk'}, & \text{(NI) [20].} \\ \frac{\pi d^2 \Delta^2 \left(\frac{2\pi e_c^2 n_{\text{int}}}{\epsilon_s} \right)^2 \exp\left(-\frac{q^2 d^2}{4}\right)}{\epsilon_q^2} u_{kk'}, & \text{(SR) [22].} \\ \frac{\left(\frac{2\pi e_c^2}{\epsilon_s q A} \right)^2 [e^{-2qb} + e^{-2q(b+l)}]}{\epsilon_q^2} u_{kk'}, & \text{(CI) [23, 32].} \\ \frac{V_a^2 \hbar q}{2A \rho_b v_{lb}} u_{kk'}, & \text{(AP) [21, 29].} \\ \frac{V_o^2 \hbar}{A \rho_b \omega_o} u_{kk'}, & \text{(OP) [21, 29].} \\ \frac{V_s e_c^2 [e^{-2qb} + e^{-2q(b+l)}]}{A q} u_{kk'}, & \text{(SPP) [21, 26].} \end{cases} \quad (7)$$

Here, $v_{kk'}$ denotes the effective interaction matrix element for the corresponding scattering mechanism, including the BLG chiral overlap factor $u_{kk'}$. The mechanism-dependent part of the matrix element contains the relevant interaction strength, momentum dependence, disorder correlations, phonon normalization, and, where appropriate, dielectric screening. The explicit forms also retain the normalization-area factors associated with the particular interaction Hamiltonians. These factors must be treated consistently when the matrix elements are inserted into the Fermi-golden-rule transition rates. In particular, the final scattering rates must have dimensions of inverse time and must be independent of the arbitrary normalization area A .

For the inelastic phonon mechanisms, the appropriate phonon energy $\hbar\omega_q$ enters the transition probability through Eq. (5), whereas for elastic mechanisms the energy-conservation condition reduces to $E_{k'} = E_k$ [29].

The quantities V_a and V_o denote the deformation-potential coupling strengths for APs and OPs, respectively, whereas V_s characterizes the coupling to substrate SPPs [26]. The SPP coupling parameter is given by

$$V_s = \frac{\hbar\omega_{sp}}{2} \left(\frac{1}{\epsilon_h + \epsilon_0} - \frac{1}{\epsilon_l + \epsilon_0} \right), \quad (8)$$

where ϵ_h and ϵ_l denote the high- and low-frequency dielectric parameters of the substrate, respectively, and ϵ_0 is the permittivity of free space.

The transport relaxation time for each scattering mechanism is evaluated independently using Eqs. (1)–(7), together with the corresponding dielectric screening function. When the different scattering mechanisms can be treated as independent, their transport scattering rates are additive according to Matthiessen's rule, $\frac{1}{\tau_{\text{tot}}(E)} = \sum_i \frac{1}{\tau_i(E)}$, where the sum extends over the scattering mechanisms considered here: NI, SR, CI, AP, OP, and SPP scattering.

The summary of the principal carrier-scattering mechanisms considered in BLG with the underlying interaction, characteristic energy scale and scattering process, relative importance, and conditions under which each mechanism can become significant has been tabulated in Table 1.

3 Neutral Impurity Scattering

In this section, we consider elastic scattering by weak finite-range NIs within the framework of the first Born approximation. Such a model can represent weak neutral disorder associated with substitutional defects, weak neutral adsorbates, impurity clusters, and smooth spatially correlated disorder. [5, 49]. The NI matrix element in Eq. (4), obtained by substituting the NI coupling function from Eq. (7), can be evaluated explicitly for a finite-range Gaussian impurity potential, yielding an analytical expression for the corresponding transport relaxation time. The impurity potential is modeled as $V(r) = V_0 \exp\left(-\frac{r^2}{2a^2}\right)$, where V_0 denotes the amplitude of the impurity potential and a characterizes its spatial extent. Finite-range Gaussian potentials are widely used as model disorder potentials to describe spatially correlated, smooth neutral disorder in graphene-based transport calculations. This model is appropriate for weak disorder, for which multiple-scattering effects are negligible and the first Born approximation is applicable. It is not intended to describe resonant scatterers, such as vacancies or hydrogen adatoms, whose strong localized potentials require a non-perturbative T -matrix treatment [40].

The two-dimensional Fourier transform of the Gaussian impurity potential is $V(q) = 2\pi a^2 V_0 \exp\left(-\frac{q^2 a^2}{2}\right)$, and hence $|V(q)|^2 = (2\pi a^2 V_0)^2 \exp(-q^2 a^2)$. The effective squared coupling function for NI is written in Eq. (7) $|v_{kk'}|^2 = (2\pi a^2 V_0)^2 e^{-q^2 a^2} u_{kk'}$. Using the NI transition probability from Eq. (7) in the general transport relaxation-rate expression of Eq. (1) and, for the elastic NI contribution in the present transport formulation, the PB factor reduces to unity. The transport relaxation rate at the Fermi energy is therefore

$$\frac{1}{\tau_{NI}(E_F)} = \frac{2\pi n_{NI} g_s g_v}{\hbar} \int \frac{d^2 k'}{(2\pi)^2} |V(q)|^2 u_{kk'} (1 - \cos \theta_{kk'}) \delta(E_k - E_{k'}), \quad (9)$$

where n_{NI} is the NI density.

For the transport rate evaluated at the Fermi energy, elastic energy conservation gives $k = k' = k_F$, and therefore $q = 2k_F \sin\left(\frac{\theta_{kk'}}{2}\right)$. Consequently, $|V(q)|^2 = (2\pi a^2 V_0)^2 \exp\left[-4k_F^2 a^2 \sin^2\left(\frac{\theta_{kk'}}{2}\right)\right]$.

Using the parabolic BLG dispersion and the chiral overlap factor [5] $u_{kk'} = \frac{1 + \cos(2\theta_{kk'})}{2} = \cos^2 \theta_{kk'}$, the scattering rate becomes

$$\frac{1}{\tau_{NI}} = \frac{2\pi n_{NI}}{\hbar} (2\pi a^2 V_0)^2 D(E_F) \frac{1}{2\pi} \int_0^{2\pi} e^{-4k_F^2 a^2 \sin^2(\theta/2)} \cos^2 \theta (1 - \cos \theta) d\theta, \quad (10)$$

where $D(E_F) = \frac{g_s g_v m^*}{2\pi \hbar^2} = \frac{2m^*}{\pi \hbar^2}$, is the total DOS including spin and valley degeneracies, with $g_s = g_v = 2$.

Scattering mechanism	Interaction	Energy scale / process	Relative importance	Conditions / representative references
Neutral impurity (NI)	Short-range disorder	Elastic; no characteristic phonon energy	Low–moderate	Important in the presence of short-range neutral defects or adsorbates; sensitive to impurity density and correlation length. [38–40]
Surface roughness (SR)	Interface disorder	Elastic; no characteristic phonon energy	Low–moderate	Relevant for rough or imperfect dielectric interfaces; depends strongly on roughness amplitude and correlation length. [22, 41]
Charged impurity (CI)	Screened Coulomb interaction	Elastic; no characteristic phonon energy	Moderate–high	Often important at low carrier density; strongly dependent on charged impurity density, dielectric environment, screening, and impurity distance. [35, 37, 42]
Acoustic phonon (AP)	Deformation potential	Quasi-elastic; $\hbar\omega_q \ll k_B T$	Moderate	Becomes increasingly important with temperature and can dominate intrinsic transport in relatively clean samples. [36, 43]
Intrinsic optical phonon (OP)	Deformation potential	Inelastic; $\hbar\omega_o$	Low below threshold; high at elevated carrier energies	Important for hot carriers and high electric fields; emission requires $E_k \geq \hbar\omega_o$. Below the emission threshold, the OP contribution is strongly suppressed by restricted inelastic phase space. [44–47]
Surface polar phonon (SPP)	Fröhlich interaction	Inelastic; $\hbar\omega_{sp}$	Moderate–high	Important in graphene supported on polar or high- κ dielectric substrates; strongly dependent on substrate phonon energy and graphene–substrate separation. [24, 25, 48]
Total	Combined scattering	Combined elastic and inelastic processes	System dependent	Total transport rate obtained from Matthiessen’s rule, $\tau_{\text{tot}}^{-1} = \sum_i \tau_i^{-1}$. The dominant mechanism depends on carrier density, temperature, substrate, disorder, and carrier energy. [35]

Table 1: Summary of the principal carrier-scattering mechanisms considered in bilayer graphene (BLG). The table specifies the underlying interaction, characteristic energy scale and scattering process, relative importance, and conditions under which each mechanism can become significant. The relative importance is qualitative and depends on carrier density, temperature, substrate, disorder strength, carrier energy, and the parameters characterizing the electron–phonon or impurity coupling. The corresponding coupling matrix elements are given in Eq. (7). The total transport scattering time is obtained from Matthiessen’s rule, $\tau_{\text{tot}}^{-1} = \sum_i \tau_i^{-1}$.

Introducing $x = 2k_F^2 a^2$, and using $e^{-4k_F^2 a^2 \sin^2(\theta_{kk'}/2)} = e^{-x} e^{x \cos \theta_{kk'}}$, together with $\cos^2 \theta_{kk'} (1 - \cos \theta_{kk'}) = \frac{1}{2} - \frac{3}{4} \cos \theta_{kk'} + \frac{1}{2} \cos 2\theta_{kk'} - \frac{1}{4} \cos 3\theta_{kk'}$, and $\int_0^{2\pi} e^{x \cos \theta_{kk'}} \cos(n\theta_{kk'}) d\theta_{kk'} = 2\pi I_n(x)$, where $I_n(x)$ is the modified Bessel function of the first kind of order n , yields

$$\frac{1}{\tau_{NI}} = \frac{2\pi n_{\text{NI}}}{\hbar} (2\pi a^2 V_0)^2 D(E_F) e^{-x} \left[\frac{1}{2} I_0(x) - \frac{3}{4} I_1(x) + \frac{1}{2} I_2(x) - \frac{1}{4} I_3(x) \right]. \quad (11)$$

In the short-range limit, the Gaussian potential approaches a contact potential provided the integrated potential strength $U_0 = 2\pi a^2 V_0$ is held constant as $a \rightarrow 0$. Since $I_0(0) = 1$, $I_n(0) = 0$, $n \geq 1$, Eq. (11)

reduces to

$$\frac{1}{\tau_{NI}^{(0)}} = \frac{\pi n_{NI}}{\hbar} U_0^2 D(E_F) = \frac{2n_{NI}m^*U_0^2}{\hbar^3}, \quad (12)$$

Thus, within the low-energy two-band approximation, the short-range NI contribution gives an energy-independent transport scattering rate, consistent with the established behavior of short-range disorder in BLG [5, 40].

4 Surface Roughness Scattering

In practical substrate-supported BLG devices, the graphene–dielectric interface is not atomically flat but exhibits random height fluctuations. These fluctuations modify the local electrostatic potential experienced by charge carriers and can therefore contribute to SR scattering [22, 41, 50]. The corresponding squared

coupling function in Eq. (7) is $|v_{kk'}^{SR}|^2 = \frac{\pi d^2 \Delta^2 \left(\frac{2\pi e_c^2 n_{\text{int}}}{\epsilon_s} \right)^2 \exp\left(-\frac{q^2 d^2}{4}\right)}{\epsilon_q^2} u_{kk'}$, as adopted in SR models for graphene-based structures [22, 41]. For elastic SR, $k' = k = k_F$. It is convenient to transform the angular integral into an integral over the momentum transfer $q = |\mathbf{k}' - \mathbf{k}| = 2k_F \sin\left(\frac{\theta_{kk'}}{2}\right)$. Consequently, $(1 - \cos \theta_{kk'}) u_{kk'} d\theta_{kk'} = \frac{q^2 dq}{2k_F^3 \sqrt{1 - q^2/(4k_F^2)}} \left[1 - \frac{q^2}{k_F^2} \left(1 - \frac{q^2}{4k_F^2} \right) \right]$. The two angular branches corresponding to $\theta_{kk'}$ and $2\pi - \theta_{kk'}$ give identical contributions. Thus, the full angular integral over $0 \leq \theta_{kk'} \leq 2\pi$ can be reduced to an integral over $0 \leq q \leq 2k_F$, with the corresponding symmetry factor included in the prefactor.

The resulting general screened SR rate is

$$\frac{1}{\tau_{\text{SR}}} = \pi d^2 \Delta^2 \left(\frac{2\pi e_c^2 n_{\text{int}}}{\epsilon_s} \right)^2 \frac{D(E_F)}{2\hbar k_F^3} \times \int_0^{2k_F} \frac{\exp\left(-\frac{q^2 d^2}{4}\right)}{\epsilon_q^2} \frac{q^2}{\sqrt{1 - \frac{q^2}{4k_F^2}}} \left[1 - \frac{q^2}{k_F^2} \left(1 - \frac{q^2}{4k_F^2} \right) \right] dq. \quad (13)$$

For the screened case, and after evaluating the momentum integral analytically, Eq. (13) reduces to

$$\frac{1}{\tau_{\text{SR}}} = \frac{\pi^2 \Delta^2 n_s^2 \hbar}{m^* d^2 D(E_F) E_k} e^{-K_s} \left(8(2K_s^2 + 2K_s + 3) Bl[0, K_s] - (4K_s^2 + 8)(2K_s + 3) {}_0F_1 \left[2, \frac{K_s^2}{4} \right] \right) \quad (14)$$

where $K_s = \frac{m^* d^2 E_k}{\hbar^2}$, $Bl[i, x]$ denote the i^{th} -order Bessel function and ${}_0F_1[a, x] = \sum_0^\infty \frac{1}{(a)_k} \frac{x^k}{k!}$ is regularised Hypergeometric function, with argument x . Equation (14) constitutes the general screened SR scattering rate within the present model. The exponential factor arises from the assumed Gaussian height–height correlation of the SR and therefore determines the momentum dependence of the roughness spectrum.

To obtain an analytical expression, we first consider the unscreened limit, $\epsilon_q = 1$. Using $q = 2k_F \sin\left(\frac{\theta_{kk'}}{2}\right)$, the Gaussian factor becomes $\exp\left[-k_F^2 d^2 \sin^2\left(\frac{\theta_{kk'}}{2}\right)\right] = e^{-x_{\text{SR}}} e^{x_{\text{SR}} \cos \theta_{kk'}}$, where $x_{\text{SR}} = \frac{k_F^2 d^2}{2}$.

Thus, the unscreened SR scattering rate becomes

$$\frac{1}{\tau_{\text{SR}}^{(0)}} = \frac{\pi d^2 \Delta^2 D(E_F)}{4\hbar} \left(\frac{2\pi e_c^2 n_{\text{int}}}{\epsilon_s} \right)^2 \int_0^{2\pi} e^{-x_{\text{SR}}} e^{x_{\text{SR}} \cos \theta_{kk'}} \cos^2 \theta_{kk'} (1 - \cos \theta_{kk'}) d\theta_{kk'}. \quad (15)$$

Using $\int_0^{2\pi} e^{x \cos \theta} \cos(n\theta) d\theta = 2\pi I_n(x)$, where $I_n(x)$ is the modified Bessel function of the first kind of order n , Eq. (15) yields

$$\frac{1}{\tau_{\text{SR}}^{(0)}} = \frac{\pi^2 d^2 \Delta^2}{2\hbar} \left(\frac{2\pi e_c^2 n_{\text{int}}}{\epsilon_s} \right)^2 D(E_F) e^{-x_{\text{SR}}} \left[\frac{1}{2} I_0(x_{\text{SR}}) - \frac{3}{4} I_1(x_{\text{SR}}) + \frac{1}{2} I_2(x_{\text{SR}}) - \frac{1}{4} I_3(x_{\text{SR}}) \right]. \quad (16)$$

In the short-correlation-length limit, $x_{\text{SR}} \rightarrow 0$, using $I_0(0) = 1$, $I_n(0) = 0$, $n \geq 1$, Eq. (16) reduces to

$$\frac{1}{\tau_{\text{SR}}^{(0)}} = \frac{\pi^2 d^2 \Delta^2}{4\hbar} \left(\frac{2\pi e_c^2 n_{\text{int}}}{\epsilon_s} \right)^2 D(E_F). \quad (17)$$

Thus, in the absence of dielectric screening, the SR scattering rate admits a closed-form analytical expression in terms of modified Bessel functions. Its functional form is analogous to that obtained for finite-range NI scattering because both mechanisms contain Gaussian momentum-dependent factors and the same BLG chiral overlap factor. Their physical origins and coupling prefactors, however, are distinct: the Gaussian factor in NI scattering originates from the finite spatial range of an individual impurity potential, whereas in SR scattering it arises from the statistical height–height correlation function of the graphene–substrate interface.

For the full momentum-dependent screening function ϵ_q , the factor ϵ_q^{-2} prevents the angular integral from reducing to the same finite combination of modified Bessel functions. The fully screened SR scattering rate is therefore evaluated numerically from Eq. (13).

5 Charged Impurity Scattering

CI scattering is an important mechanism limiting carrier mobility in BLG, particularly in devices supported on dielectric substrates such as SiO_2 . [23, 32] CIs arising from substrate defects, adsorbed ions, or fabrication-induced contaminants generate long-range Coulomb potentials that scatter charge carriers and degrade electronic transport. Screened CI scattering has been extensively studied in graphene transport, particularly in MLG. [51, 52] For BLG, the corresponding transport rate incorporates the parabolic low-energy dispersion, the BLG chiral overlap factor, and dielectric screening.[32]

For CI scattering, the mechanism-specific matrix element is obtained from the screened Coulomb interaction together with the distance-dependent form factor associated with the impurity location.[23, 32] Using this matrix element in the general transport-rate expression Eq. (1) gives

$$\frac{1}{\tau_{\text{CI}}} = \frac{2m^* n_{\text{CI}} g_s g_v}{\pi \hbar^3} \int_0^{2\pi} \left(\frac{2\pi e_c^2}{\epsilon_s q} \right)^2 \left(1 + \frac{4m^* e_c^2}{\epsilon_s \hbar^2 q} \right)^{-2} \left[e^{-2qb} + e^{-2q(b+l)} \right] \frac{1 + \cos 2\theta_{kk'}}{2} (1 - \cos \theta_{kk'}) d\theta_{kk'}, \quad (18)$$

where n_{CI} is the CI density, $q = 2k \sin \left(\frac{\theta_{kk'}}{2} \right)$, $E_{k'} = E_k$, and the PB ratio is unity for elastic scattering. The Thomas–Fermi screening wave vector is $q_{\text{TF}} = \frac{4m^* e_c^2}{\epsilon_s \hbar^2}$.

Here the two exponential terms represent the contributions associated with CIs located at distances b and $b+l$ from the reference BLG layer. The expression assumes statistically independent contributions from the two impurity populations. If coherent interference between the corresponding scattering amplitudes is included, the distance-dependent form factor must instead be modified to contain the appropriate cross term.

The angular integral in Eq. (18) is difficult to evaluate analytically in its general form. But under strong screening, $\alpha = \frac{q_{\text{TF}}}{2k} \gg 1$, the resulting integral Eq. (18) can be evaluated analytically, giving

$$\begin{aligned} \frac{1}{\tau_{\text{CI}}} &= \mathcal{A}(\gamma) + \mathcal{A}(\Gamma) \\ &\text{where} \\ \mathcal{A}(\gamma) &= -\frac{1}{48\gamma^4 m^*} \pi n_{\text{CI}} \hbar \left(-32\gamma^3(3+4\gamma^2) + 9\pi\gamma(5+4\gamma^2) (Sl[1, 4\gamma] - Bl[1, 4\gamma]) + 3\pi(15+25\gamma^2+16\gamma^4) (Bl[2, 4\gamma] \right. \\ &\quad \left. - Sl[2, 4\gamma]) \right) \end{aligned} \quad (19)$$

Where $\gamma = (d/\hbar)\sqrt{2m^*E_k}$, $\Gamma = ((c+d)/\hbar)\sqrt{2m^*E_k}$, where $Bl[i, x]$ and $Sl[i, x]$ denote the i^{th} -order Bessel and Struve functions, respectively, with argument x . Equation (19) provides the analytical CI relaxation rate. For the parabolic BLG dispersion, $k = \frac{\sqrt{2m^*E_k}}{\hbar}$, so that $\alpha \propto E_k^{-1/2}$.

If screening is removed then the Eq. (18) yields the following analytical result

$$\frac{\pi^2 e_c^4 n_{CI}}{16 E_k \epsilon_s^2 \hbar} (\mathcal{B}(\gamma) + \mathcal{B}(\Gamma))$$

where

$$\mathcal{B}(\gamma) = \frac{1}{\gamma^3} (-24\gamma \text{Sl}[2, 4\gamma] + (64\gamma^3 + 24\gamma) \text{Bl}[0, 4\gamma] - (32\gamma^2 + 12) \text{Bl}[1, 4\gamma] + 32\gamma^2 \text{Sl}[1, 4\gamma] - 32\gamma^3 \text{Sl}[0, 4\gamma]) \quad (20)$$

A particularly simple analytical result follows if $b = l = 0$ in the exponential factors in Eq. (18) and in the strong-screening regime, where $q_{TF} \gg q$. In this limit, $\left(1 + \frac{q_{TF}}{q}\right)^{-2} \simeq \frac{q^2}{q_{TF}^2}$, and hence

$$\frac{1}{\tau_{CI}^{(0)}} \simeq \frac{4\pi m^* n_{CI} e_c^4}{\epsilon_s^2 \hbar^3 q_{TF}^2} \int_0^{2\pi} \cos^2 \theta_{kk'} (1 - \cos \theta_{kk'}) d\theta_{kk'} = \frac{4\pi m^* n_{CI} e_c^4}{\epsilon_s^2 \hbar^3 q_{TF}^2} \pi = \frac{\pi^2}{4} \frac{\hbar n_{CI}}{m^*}.$$

Thus, within the strong-screening and zero-separation approximations, the CI scattering rate becomes independent of the carrier wave vector k to leading order. The explicit dependence on the dielectric constant and Coulomb interaction strength cancels through the Thomas–Fermi screening wave vector, leaving a rate proportional to the CI density n_{CI} and inversely proportional to the effective mass m^* .

6 Acoustic Phonon Scattering

AP constitute an important intrinsic scattering mechanism in BLG, particularly at finite lattice temperature.[29, 36] Here we consider longitudinal APs and evaluate the energy-dependent carrier relaxation rate both with and without PB. The analytical results are subsequently compared with numerical calculations.

For longitudinal APs, a linear dispersion relation is assumed, $\omega_q = v_{lb} q$, where v_{lb} is the longitudinal AP velocity and q is the phonon wave vector.[29, 36]

Using the acoustic deformation-potential coupling from Eq. (7), $\frac{V_a^2 \hbar q}{2A \rho_b v_{lb}} u_{kk'}$, and the BLG chiral overlap factor,[5, 32] the transport relaxation rate Eq. (1), including PB (with $f(x) = (1 + \exp((x - \mu)/K_B T))^{-1}$, $\mu \rightarrow$ chemical potential), is written as

$$\begin{aligned} \frac{1}{\tau_{AP}^{PB}} = & \frac{V_a^2 m^*}{4\pi \hbar^2 \rho_b v_{lb}^2} \int_0^\infty \int_0^{2\pi} \frac{1 - f(E_{k'})}{1 - f(E_k)} \\ & \times \omega_{ba} [n_{ph} \delta(E_k + \hbar \omega_q - E_{k'}) + (n_{ph} + 1) \delta(E_k - \hbar \omega_q - E_{k'})] \\ & \times \frac{1 + \cos(2\theta_{kk'})}{2} (1 - \cos \theta_{kk'}) d\theta_{kk'} dE_{k'}. \end{aligned} \quad (21)$$

The resulting expression follows from the Fermi–Dirac occupation factors and the AP transition probabilities within the Boltzmann transport framework.[29, 32] Evaluating the energy integral and carrying out the angular integration under the approximations adopted here gives

$$\frac{1}{\tau_{AP}^{PB}} = V_a^2 m^* \omega_{ba} \frac{n_{ph}(1 - f(E_k + \hbar \omega_{ba})) + (n_{ph} + 1)(1 - f(E_k - \hbar \omega_{ba}))\theta(E_k - \hbar \omega_{ba})}{4\rho_b v_{lb}^2 \hbar^2 (1 - f(E_k))}. \quad (22)$$

Here $\omega_{ba} = \frac{4\sqrt{2m^* E_k} v_{lb}}{\pi \hbar}$, $n_{ph} = \frac{1}{\exp(\hbar \omega_{ba}/(K_B T)) - 1}$ and $\Theta(x)$ denotes the Heaviside step function. Its appearance in the emission contribution imposes the kinematic threshold required for phonon emission.

The absence of the PB factor considerably simplifies the analytical evaluation. The corresponding relaxation rate without PB is

$$\frac{1}{\tau_{AP}} = V_a^2 m^* \omega_{ba} \frac{n_{ph} + (n_{ph} + 1)\theta(E_k - \hbar \omega_{ba})}{4\rho_b v_{lb}^2 \hbar^2}. \quad (23)$$

Equations (22) and (23) provide the analytical AP relaxation rates with and without PB, respectively. Their comparison isolates the effect of the electronic occupation factors on AP scattering. The temperature dependence arises from both the phonon occupation factors and, when PB is retained, the Fermi–Dirac occupation factors. Increasing the lattice temperature generally enhances the AP population and therefore increases the available scattering phase space.

7 Optical Phonon Scattering

OP scattering provides an inelastic relaxation channel whose importance depends strongly on the carrier-energy and temperature regime and on the particular BLG phonon branches included.[44, 53] In contrast to APs, OPs have a finite characteristic energy and arise from out-of-phase lattice vibrations. Within the present model, the OP branch is treated as approximately dispersionless over the relevant range of wave vectors, so that $\omega_q \simeq \omega_o$, where ω_o is the OP frequency.

The OP occupation is described by the Bose–Einstein distribution, $N_o = \frac{1}{\exp(\beta\hbar\omega_o)-1}$. Using the optical deformation-potential matrix element specified in Eq. (7), $|v_{kq}|^2 = \frac{V_o^2\hbar}{A\rho_b\omega_o} u_{kk'}$, [44] together with the BLG chiral overlap factor, the transport relaxation rate in Eq. (1) gives the PB-inclusive OP scattering rate

$$\begin{aligned} \frac{1}{\tau_{\text{OP}}^{\text{PB}}} = & \frac{V_o^2 m^*}{2\pi\hbar^2 \rho_b \omega_o} \int_0^\infty \int_0^{2\pi} \frac{1-f(E_{k'})}{1-f(E_k)} \\ & \times [N_o \delta(E_k + \hbar\omega_o - E_{k'}) + (N_o + 1) \delta(E_k - \hbar\omega_o - E_{k'})] \\ & \times \frac{1 + \cos(2\theta_{kk'})}{2} (1 - \cos\theta_{kk'}) d\theta_{kk'} dE_{k'}. \end{aligned} \quad (24)$$

Carrying out the angular integration using $\int_0^{2\pi} \frac{1+\cos(2\theta)}{2} (1 - \cos\theta) d\theta = \pi$, and using $k'dk' = \frac{m^*}{\hbar^2} dE_{k'}$, the energy integration over the absorption and emission delta functions yields

$$\frac{1}{\tau_{\text{OP}}^{\text{PB}}} = \frac{V_o^2 m^*}{2\rho_b \omega_o \hbar^2} \left[N_o \frac{1-f(E_k + \hbar\omega_o)}{1-f(E_k)} + (N_o + 1) \Theta(E_k - \hbar\omega_o) \frac{1-f(E_k - \hbar\omega_o)}{1-f(E_k)} \right]. \quad (25)$$

To isolate the effect of PB, the corresponding rate is also evaluated by setting the occupation-factor ratio to unity. The corresponding analytical expression is

$$\frac{1}{\tau_{\text{OP}}} = \frac{V_o^2 m^* \left[e^{\frac{\hbar\omega_o}{k_B T}} \Theta(E_k - \hbar\omega_o) + 1 \right]}{2\rho_b \omega_o \hbar^2 \left(e^{\frac{\hbar\omega_o}{k_B T}} - 1 \right)}. \quad (26)$$

Equation (26) shows that, in the absence of PB, the OP scattering rate is determined by the phonon occupation and the kinematic threshold for phonon emission.[53] The factor $\Theta(E_k - \hbar\omega_o)$ ensures that emission is allowed only when $E_k \geq \hbar\omega_o$, whereas absorption remains possible at all carrier energies provided that thermally populated OPs are available. Thus, within the present dispersionless single-mode approximation, OP scattering possesses a finite-energy emission threshold set by $\hbar\omega_o$.

8 Surface Polar Phonon Scattering

SPPs are optical vibrational modes localized near the surface of polar dielectric substrates. Their relative displacement of positively and negatively charged ions generates macroscopic electric fields that couple to charge carriers through the long-range Fröhlich interaction. Because these evanescent electric fields penetrate into the adjacent graphene layers, SPP scattering provides an important channel for carrier momentum and energy relaxation, particularly at elevated temperatures.[24–26]

Electron scattering by SPPs has been extensively investigated in MLG on polar substrates [29]. Although the underlying Boltzmann transport formalism is unchanged, its application to BLG requires the appropriate parabolic dispersion, DOS, chiral overlap factor, and dielectric environment to be taken into account.[21, 32]

Assuming that the carrier density is equally distributed between the two graphene layers, the SPP transport relaxation rate is written as

$$\frac{1}{\tau_{SP}^{PB}} = \frac{m^* V_s e_c^2}{2\pi\hbar^3} \int_0^\infty \int_0^{2\pi} \frac{1 - f(E_{k'})}{1 - f(E_k)} \frac{e^{-2qb} + e^{-2q(b+l)}}{2q} \times [n_{ph}\delta(E_k + \hbar\omega_s - E_{k'}) + (n_{ph} + 1)\delta(E_k - \hbar\omega_s - E_{k'})] \times \frac{1 + \cos(2\theta_{kk'})}{2} (1 - \cos\theta) d\theta_{kk'} dE_{k'}, \quad (27)$$

where $n_{ph} = \frac{1}{\exp(\hbar\omega_s/k_B T) - 1}$ is the Bose-Einstein occupation number of the SPP mode, ω_s is the SPP frequency, and b and $b+l$ characterize the distances of the two graphene layers from the dielectric interface.

The first and second terms in Eq. (27) describe phonon absorption and emission, respectively. In particular, the emission process is kinematically allowed only when $E_k \geq \hbar\omega_s$.

Performing the energy and angular integrations analytically gives

$$\frac{1}{\tau_{SP}^{PB}} = \frac{m^* V_s e_c^2}{\sqrt{2}\pi\hbar^2} [n_{ph}T_{f+} + (n_{ph} + 1)T_{f-}\Theta(E_k - \hbar\omega_s)], \quad (28)$$

where the functions T_{f+} and T_{f-} contain the carrier-energy dependence associated with phonon absorption and emission, respectively, including the PB factor.

$$T_{f\pm} = \frac{\mathcal{F}_\pm}{30\sqrt{2}\pi\alpha_\pm^3\hbar^2\sqrt{(2\alpha_\pm + \beta_\pm)}} \times \left[\sqrt{\beta_\pm^2 - 4\alpha_\pm^2} (-17\alpha_\pm^2 - 2\hbar^2\omega_s^2 + 5\alpha_\pm\beta_\pm)E[A_\pm] - (14E_k\alpha_\pm^2 + 11E_k\hbar^2\omega_s^2 \pm \hbar\omega_s(7E_k^2 + 2\hbar^2\omega_s^2) + 14\alpha_\pm^3 - \alpha_\pm\hbar^2\omega_s^2)E[B_\pm] + \sqrt{\beta_\pm^2 - 4\alpha_\pm^2} (13\alpha_\pm^2 + 2\hbar^2\omega_s^2 - 2E_k\alpha_\pm - \alpha_\pm\hbar\omega_s)K[A_\pm] + (30E_k^3 + 45E_k^2\hbar\omega_s + 19E_k\hbar^2\omega_s^2 + 2\hbar^3\omega_s^3 - 30\alpha_\pm^3 - 5\alpha_\pm\hbar^2\omega_s^2)K[B_\pm] \right]. \quad (29)$$

Here, $E(\cdot)$ and $K(\cdot)$ denote the complete elliptic integrals of the second and first kinds, respectively, with arguments $A_\pm = \frac{-4\sqrt{E_k(E_k \pm \hbar\omega_s)}}{2E_k \pm \hbar\omega_s - 2\sqrt{E_k(E_k \pm \hbar\omega_s)}}$, $B_\pm = \frac{4\sqrt{E_k(E_k \pm \hbar\omega_s)}}{2E_k \pm \hbar\omega_s + 2\sqrt{E_k(E_k \pm \hbar\omega_s)}}$, $\alpha_\pm = \sqrt{E_k(E_k \pm \hbar\omega_s)}$, $\beta_\pm = 2E_k \pm \hbar\omega_s$, and $\mathcal{F}_\pm = \frac{1 - F_{db}(E_k \pm \hbar\omega_s)}{1 - F_{db}(E_k)}$.

For the emission contribution, the expressions containing the minus sign are understood only for $E_k \geq \hbar\omega_s$, or equivalently are multiplied by $\Theta(E_k - \hbar\omega_s)$.

To isolate the effect of PB, we also consider the approximation $\frac{1-f(E_{k'})}{1-f(E_k)} \simeq 1$. Under this approximation, Eq. (27) becomes

$$\frac{1}{\tau_{SP}} = \frac{V_s e_c^2 m^*}{\sqrt{2}\pi\hbar^2} [n_{ph}T_+ + (n_{ph} + 1)T_-\Theta(E_k - \hbar\omega_s)], \quad (30)$$

where $T_\pm$ are obtained from $T_{f\pm}$ by removing the PB ratio ($\mathcal{F}_\pm = 1$). The emission term remains subject to the kinematic condition $E_k \geq \hbar\omega_s$.

9 Results and Discussion

The transport relaxation rate is one of the key quantities governing electronic transport in BLG. It is strongly influenced by various scattering mechanisms, including NI, SR, CI, AP, OP, and SPP scatterings. Understanding the relative contributions of these mechanisms is essential for assessing and optimizing the transport performance of BLG-based electronic devices.[21, 23, 24, 26, 32, 36]

In this section, the energy-dependent transport relaxation rates corresponding to the individual scattering mechanisms are presented and discussed. Unless stated otherwise, the carrier energy is varied over the range

$0 \leq E_k \leq 0.35$ eV. The analytical expressions derived in the preceding sections are evaluated numerically, and the resulting rates are verified by direct numerical evaluation of the corresponding transport integrals.

The principal parameters used in the numerical calculations are summarized in Table 2. The effective mass follows the standard low-energy effective-mass description of Bernal-stacked BLG.[5]

Symbol	Physical meaning	Value
$\hbar$	Reduced Planck constant	1.05457×10^{-27} erg s
A	Area of the BLG sheet	Defined by sample geometry
m^*	Effective mass of electrons in BLG	$0.033 m_e$
k_F	Fermi wave vector	Calculated from n
n	Carrier concentration	$1 \times 10^{12} \text{ cm}^{-2}$
e_c	Elementary charge in Gaussian units	4.8032×10^{-10} esu
k_B	Boltzmann constant	1.38065×10^{-16} erg K $^{-1}$
ρ_b	Areal mass density of BLG	$15.2 \times 10^{-8} \text{ g cm}^{-2}$
v_{lb}	Longitudinal acoustic phonon velocity	$2.12 \times 10^6 \text{ cm s}^{-1}$ [36]
V_0	Neutral impurity scattering potential	1.0 eV
a	Neutral impurity correlation length	0.5 nm
n_{CI}	Charged impurity concentration	$1 \times 10^{10} \text{ cm}^{-2}$
n_{int}	Interface charge density	$1 \times 10^{10} \text{ cm}^{-2}$
V_a	Acoustic deformation potential	19 eV
V_o	Optical deformation potential	$2.5 \text{ eV \AA}^{-1}$
V_s	Surface polar phonon coupling parameter	Given by Eq. (8)
ω_q	Acoustic phonon dispersion	$\omega_q = v_{lb}q$
$\hbar\omega_o$	Optical phonon energy	200 meV
$\hbar\omega_{sp}$	Surface polar phonon energy	59.8 meV (SiO $_2$ substrate)
ϵ_s	Effective background dielectric constant	2.45
ϵ_l	Low-frequency relative dielectric constant	3.9
ϵ_h	High-frequency relative dielectric constant	2.5
ϵ_q	Static dielectric screening function	Given by Eq. (6)
b	Distance between the substrate and BLG	50 \AA
l	Interlayer spacing of BLG	3.37 \AA
d	Surface-roughness correlation length	100 \AA
h	Surface-roughness amplitude	5 \AA
T	Temperature used in the calculations	150 K (AP); 200 K (OP and SPP)

Table 2: Material parameters and symbols used in the scattering formalism. Numerical parameters correspond to the values adopted in the present calculations. Parameters without numerical values are defined symbolically and evaluated through the corresponding equations.

9.1 Neutral Impurity Scattering

Unless otherwise stated, the numerical calculations for NI scattering use a NI density $n_{\text{NI}} = 10^{11} \text{ cm}^{-2}$, a Gaussian correlation length $a = 0.5$ nm, and an impurity potential amplitude $V_0 = 1.0$ eV. The parameters a and V_0 are phenomenological quantities used to characterize the spatial extent and strength of finite-range neutral disorder, respectively. Finite-range disorder models are commonly used in theoretical treatments of graphene transport.[38, 39] The values adopted here are representative simulation parameters rather than unique material constants.

Using Eq. (11), the energy dependence of the NI transport relaxation rate is evaluated numerically. The result is compared with a direct numerical evaluation of the angular integral in Eq. (10) to verify the analytical Bessel-function expression.

To place the analytical BLG result in a broader context, we compare finite-range NI scattering in BLG with the corresponding results for MLG and a conventional 2DEG. Within the same first Born approximation and Gaussian impurity model, the differences among these systems arise from their electronic band structures, DOS, and angular overlap factors.[5, 54, 55]

For all three systems, the transport relaxation rate can be written in the unified form

$$\frac{1}{\tau_{NI}} = \frac{2\pi n_{NI}}{\hbar} (2\pi a^2 V_0)^2 D(E_F) R(x), \quad x = 2k_F^2 a^2, \quad (31)$$

where $D(E_F)$ is the DOS appropriate to the system under consideration and $R(x)$ is a dimensionless transport kernel containing the angular dependence of the scattering process.

The three systems differ in their angular overlap factors. For a conventional 2DEG, $u_{2DEG} = 1$, whereas for MLG, $u_{MLG} = \frac{1+\cos\theta}{2}$. These different overlap factors lead to distinct angular weightings of the transport scattering probability.[1, 5, 55]

$$\begin{aligned} R_{2DEG}(x) &= e^{-x} [I_0(x) - I_1(x)], \\ R_{MLG}(x) &= \frac{1}{4} e^{-x} [I_0(x) - I_2(x)], \\ R_{BLG}(x) &= e^{-x} \left[\frac{1}{2} I_0(x) - \frac{3}{4} I_1(x) + \frac{1}{2} I_2(x) - \frac{1}{4} I_3(x) \right]. \end{aligned} \quad (32)$$

For MLG, Eq. (31) using (32) gives

$$\frac{1}{\tau_{NI}^{MLG}} = \frac{2\pi n_{NI}}{\hbar} (2\pi a^2 V_0)^2 D_{MLG}(E_F) e^{-x} \frac{1}{4} [I_0(x) - I_2(x)]. \quad (33)$$

Using the modified-Bessel-function identity $I_0(x) - I_2(x) = \frac{2I_1(x)}{x}$, this may equivalently be written as

$$\frac{1}{\tau_{NI}^{MLG}} = \frac{2\pi n_{NI}}{\hbar} (2\pi a^2 V_0)^2 D_{MLG}(E_F) e^{-x} \frac{I_1(x)}{2x}. \quad (34)$$

This form is consistent with the general Boltzmann transport treatment of disorder in graphene.[55] An additional distinction among the three systems is the energy dependence of the DOS. For a conventional parabolic 2DEG, $D_{2DEG}(E) = \frac{g_s m_{2DEG}^*}{2\pi\hbar^2}$, whereas for low-energy BLG, $D_{BLG}(E) = \frac{g_s g_v m_{BLG}^*}{2\pi\hbar^2}$. Thus, both the conventional 2DEG and low-energy BLG possess energy-independent DOS, although their numerical values depend on their respective effective masses and degeneracy factors. In contrast, the MLG DOS varies linearly with energy.[1, 5, 54]

In the short-range limit, $x \rightarrow 0$, the transport kernels reduce to $R_{2DEG}(0) = 1$, $R_{BLG}(0) = \frac{1}{2}$, $R_{MLG}(0) = \frac{1}{4}$. These differences reflect the distinct angular weighting introduced by the overlap factors of the three systems. The comparison between the 2DEG and BLG is particularly useful because both possess parabolic low-energy dispersions and energy-independent DOS, whereas their angular scattering structures differ because BLG carries a nontrivial pseudospin chirality. MLG introduces the additional distinction of a linear Dirac dispersion and an energy-dependent DOS.

Thus, Eq. (32) provides a compact unified description of finite-range NI scattering in 2DEG, MLG, and BLG. The 2DEG and MLG transport descriptions are consistent with established Boltzmann treatments,[54, 55] while the BLG kernel follows directly from the low-energy two-band chiral overlap factor and the analytical Gaussian-impurity calculation developed in the present work.

The energy dependence of the calculated scattering rates is shown in Fig. 1. Panel (a) compares the SR, NI, and CI scattering rates in MLG and BLG, whereas panel (b) examines the effect of dielectric screening on CI and SR scattering in BLG. The subscripts m and b denote monolayer and bilayer graphene, respectively. Figure 1(a) shows a pronounced difference between the NI scattering rates of BLG and MLG. For BLG, the NI rate, NI_b , is the dominant scattering contribution over most of the investigated energy range, although the SR contribution is larger in the very-low-energy regime. The rate NI_b is of the order of 10^{12} – 10^{13} s⁻¹ and exhibits only a weak dependence on carrier energy. This behavior is consistent with the approximately energy-independent DOS of low-energy BLG. Thus, within the present low-energy approximation, the energy

dependence of the BLG NI scattering rate arises primarily through the finite-range transport kernel, since the BLG DOS is approximately energy independent.

In contrast, the MLG NI rate, NI_m , is substantially smaller at low carrier energies and increases gradually with increasing E_k , reaching approximately 10^{10} – 10^{11} s⁻¹ toward the upper part of the investigated energy range. This behavior reflects the combined influence of the linearly increasing MLG DOS and the energy dependence of the finite-range transport kernel. Consequently, the calculated NI rates exhibit markedly different energy dependencies in MLG and BLG.

The results shown in Fig. 1(a) therefore provide a clear numerical manifestation of the different electronic structures of MLG and BLG and their influence on NI-limited carrier transport.

It should be noted that a separately screened NI scattering case has not been considered. Unlike CI, NI do not generate a long-range Coulomb potential, and their scattering potential is predominantly short-ranged. Hence, the long-range dielectric screening treatment used for CI scattering does not lead to an analogous screening correction for the NI potential adopted here.

9.2 Surface Roughness Scattering

The SR model adopted here is based on the interface-roughness framework discussed in Refs. [22, 41], with the corresponding formulation generalized in the present work to the low-energy BLG electronic structure. The complete screened SR scattering rate is given by Eq. (13).

To verify the analytical result, the corresponding unscreened transport integral was also evaluated numerically using Eq. (15). The analytical and direct numerical results show close agreement over the investigated energy range, confirming that the closed-form expression in Eq. (16) correctly reproduces the corresponding transport integral. The analytical unscreened SR scattering rate obtained from Eq. (16) is shown in Fig. 1(a) as a function of carrier energy. For BLG, the SR scattering rate decreases strongly with increasing carrier energy, with the most pronounced variation occurring in the low-energy region below approximately 0.1 eV. At higher carrier energies, the rate decreases more gradually, where the impurity scattering contributions, NI and CI become comparatively more important. This behavior results from the combined energy dependence of the Fermi wave vector, the Gaussian roughness factor, and the transport angular kernel. In particular, increasing carrier energy increases k_F , thereby modifying the dimensionless parameter $x_{\text{SR}} = \frac{k_F^2 d^2}{2}$, and consequently the Bessel-function combination appearing in Eq. (16). Thus, the energy dependence arises from the complete analytical expression rather than from the Gaussian roughness factor alone.

The effect of dielectric screening on SR scattering is examined in Fig. 1(b), which compares the SR rates obtained without screening, with the full momentum-dependent screening, and within the asymptotic strong-screening approximation. In the present model, screening enters the SR scattering potential through the electrostatic response of the graphene-dielectric interface. The fully screened result is obtained from Eq. (13), whereas the unscreened result corresponds to Eq. (16). The screening factor, appearing through ϵ_q^{-2} , modifies the effective scattering potential and therefore the magnitude and energy dependence of the SR relaxation rate.

As shown in Fig. 1(b), the unscreened SR rate, SR_w , is substantially larger than the screened SR rates over most of the investigated energy range. The fully screened rate, SR_f , exhibits a pronounced low-energy maximum before decreasing with increasing carrier energy. In contrast, the strong-screening approximation, SR_s , substantially suppresses the SR rate and produces a much smaller rate over the investigated range. Thus, the treatment of dielectric screening has a significant influence not only on the magnitude of SR scattering but also on its energy dependence.

A direct comparison of SR and CI scattering in Fig. 1(b) further illustrates the role of screening in BLG. In the absence of screening, both mechanisms produce relatively large relaxation rates, with the unscreened SR rate becoming particularly prominent toward the higher-energy region. When full screening is included, the SR rate exhibits a low-energy maximum and subsequently decreases, while the screened CI rate remains comparatively large over a broader energy range. Under the strong-screening approximation, the SR rate is strongly suppressed and remains below the corresponding screened CI rate over most of the investigated energy range.

These results demonstrate that the relative importance of SR and CI scattering in BLG depends sensitively on both carrier energy and the treatment of dielectric screening. In particular, screening changes the balance between the two mechanisms rather than producing a uniform rescaling of their scattering rates. The

calculated results therefore indicate that dielectric screening should be included when assessing the relative contributions of SR and CI scattering in BLG.

9.3 Charged Impurity Scattering

CI scattering is an important mechanism that contributes to the degradation of carrier mobility and conductivity in BLG. CIs associated with substrate defects, adsorbed ions, and fabrication-induced contaminants generate long-range Coulomb potentials and can therefore provide a significant source of momentum relaxation.[23, 32, 55]

Under the approximation $e^{-2qb} \simeq e^{-2q(b+l)} \simeq 1$, corresponding to impurity distances satisfying $qb \ll 1$ and $q(b+l) \ll 1$, the analytical CI relaxation rate is obtained as given in Eq. (19).

The complete expression in Eq. (18), without the zero-separation approximation, is evaluated numerically. This provides a direct assessment of the validity and range of applicability of the analytical approximation. The analytical and numerical results show similar energy dependence and are found to be in close agreement. The remaining differences in magnitude arise primarily from the treatment of the finite-distance impurity factors.

The calculated CI relaxation rate exhibits a stronger energy dependence in the low-energy region. This behavior is associated with the energy dependence of the carrier wave vector and the Thomas–Fermi screening parameter. In low-energy BLG, the parabolic dispersion $E_k = \frac{\hbar^2 k^2}{2m^*}$ gives $k \propto E_k^{1/2}$, whereas the Thomas–Fermi screening wave vector q_{TF} is approximately energy independent. Consequently, $\alpha = \frac{q_{\text{TF}}}{2k} \propto E_k^{-1/2}$. Thus, screening becomes increasingly important at low carrier energies. When the numerical calculation is performed using discrete energy points, the apparent discreteness of the plotted rate reflects the chosen energy sampling and should not be interpreted as evidence that only a limited number of physical electronic states satisfy energy conservation.

The dependence of the CI relaxation rate on impurity density is examined numerically. Within the independent-scatterer approximation, the transport rate is proportional to the CI density, $\frac{1}{\tau_{\text{CI}}} \propto n_{\text{CI}}$. This proportionality is the standard result for independent charged scatterers in transport theory.[23, 35]

The CI scattering characteristics of BLG can be placed in a broader context by comparing them with those of MLG and a conventional 2DEG. The three systems differ in their low-energy dispersions, DOS, screening properties, and wave-function overlap factors.[1, 5, 35, 54]

Within the Thomas–Fermi approximation, the effective coupling function for screened Coulomb scattering in MLG is $v_{kk'}^{\text{MLG}} = \frac{2\pi e_c^2}{\epsilon_s(q+q_{\text{TF}}^{\text{MLG}})} e^{-qb} u_{kk'}^{\text{MLG}}$, where $q_{\text{TF}}^{\text{MLG}} = 4r_s k_F$, $r_s = \frac{e_c^2}{\epsilon_s \hbar v_F}$. Thus, $q_{\text{TF}}^{\text{MLG}} \propto k_F \propto \sqrt{n}$, while $\frac{q_{\text{TF}}^{\text{MLG}}}{k_F} = 4r_s$ is independent of carrier density for a fixed dielectric environment. The result $q_{\text{TF}}^{\text{MLG}} = 4r_s k_F$ is standard for doped MLG.[35, 37]

For low-energy BLG, the Thomas–Fermi screening wave vector is $q_{\text{TF}}^{\text{BLG}} = \frac{4m^* e_c^2}{\epsilon_s \hbar^2}$, for spin and valley degeneracy $g_s g_v = 4$. [5, 32] Thus, within the parabolic low-energy approximation, $q_{\text{TF}}^{\text{BLG}}$ is independent of carrier density.

For a conventional parabolic 2DEG with total degeneracy g , $q_{\text{TF}}^{2\text{DEG}} = \frac{gm^* e_c^2}{\epsilon_s \hbar^2}$, which is likewise independent of carrier density within the ideal parabolic-band approximation. For a spin-degenerate 2DEG, $g = 2$. [54, 56]

The principal differences among the three systems are summarized in Table 3. The electronic structures and chiral overlap factors of MLG and BLG are well established, [1, 5] whereas the corresponding parabolic 2DEG results follow the standard two-dimensional electron-gas treatment.[54]

The comparison shows that low-energy BLG shares the parabolic dispersion, approximately constant DOS, and density-independent Thomas–Fermi screening of a conventional 2DEG, while retaining the chiral pseudospin structure of graphene. In particular, $u_{\text{MLG}}(\pi) = 0$, $u_{\text{BLG}}(\pi) = 1$, $u_{2\text{DEG}}(\pi) = 1$. Thus, MLG suppresses exact backscattering, whereas low-energy BLG does not. Nevertheless, the angular dependence of BLG remains distinct from that of a conventional 2DEG because of its chiral overlap factor. [1, 5]

For elastic scattering at the Fermi surface, the CI relaxation rate in MLG is

$$\frac{1}{\tau_{\text{CI}}^{\text{MLG}}} = \frac{4\pi n_{\text{CI}} r_s^2 v_F}{k_F} \int_0^1 \frac{x^2 \sqrt{1-x^2}}{(x+2r_s)^2} e^{-4k_F b x} dx, \quad (35)$$

where $x = \sin \frac{\theta}{2}$ and $q = 2k_F x$.

For finite impurity distance b , the factor $e^{-4k_F b x}$ prevents the integral from reducing to the simple closed form obtained in the $b = 0$ limit. Although formal representations in terms of special functions or convergent series can be constructed, they are less convenient for numerical evaluation. Therefore, the exact finite- b integral in Eq. (35) is evaluated numerically, while the $b = 0$ limit is treated analytically.

For impurities located in the graphene plane ($b = 0$), the remaining integral can be evaluated analytically. For $r_s > 1/2$, one obtains

$$\frac{1}{\tau_{\text{CI}}^{\text{MLG}}} = \frac{4\pi n_{\text{CI}} r_s^2 v_F}{k_F} \left[\frac{\pi}{4} + 6r_s - 6\pi r_s^2 + \frac{8r_s(6r_s^2 - 1)}{\sqrt{4r_s^2 - 1}} \left\{ \arctan \left(\frac{2r_s + 1}{\sqrt{4r_s^2 - 1}} \right) - \arctan \left(\frac{1}{\sqrt{4r_s^2 - 1}} \right) \right\} \right]. \quad (36)$$

For impurities located in the 2DEG plane ($b = 0$), the corresponding integral can also be evaluated analytically. For $r_s^{\text{2DEG}} > 1$,

$$\frac{1}{\tau_{\text{CI}}^{\text{2DEG}}} = \frac{4\pi m_{\text{2DEG}}^* n_{\text{CI}} e^4}{\hbar^3 \epsilon_s^2 k_F^2} \left[\frac{\pi}{2} - \frac{r_s^{\text{2DEG}}}{(r_s^{\text{2DEG}})^2 - 1} + \frac{r_s^{\text{2DEG}} [2 - (r_s^{\text{2DEG}})^2]}{[(r_s^{\text{2DEG}})^2 - 1]^{3/2}} \arccos \left(\frac{1}{r_s^{\text{2DEG}}} \right) \right]. \quad (37)$$

For finite impurity distance b , the corresponding integral is evaluated numerically using the 2DEG DOS. Unlike MLG and BLG, no chiral wave-function overlap factor appears in the conventional 2DEG transport integral.

Overall, CI scattering in BLG combines the approximately density-independent screening characteristic of a parabolic two-dimensional system with a nontrivial graphene pseudospin overlap, resulting in transport characteristics distinct from those of both MLG and a conventional 2DEG. [5, 32, 54]

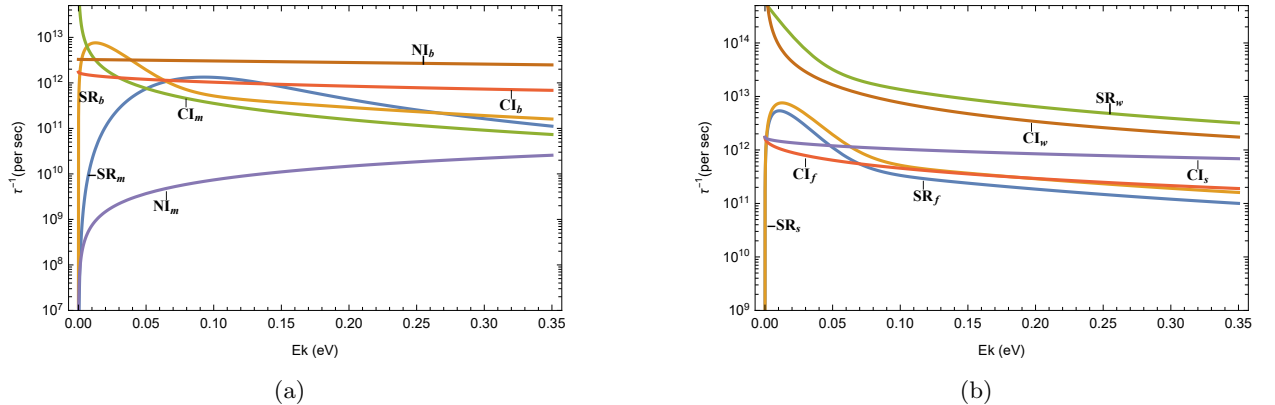


Figure 1: (a) Energy dependence of the carrier scattering rates, τ^{-1} , in monolayer (subscript m) and bilayer (subscript b) graphene due to surface-roughness (SR), neutral-impurity (NI), and charged-impurity (CI) scattering. (b) Energy dependence of the CI and SR scattering rates in bilayer graphene for three screening regimes: without screening (w), full screening (f), and the asymptotic strong-screening approximation (s). The corresponding rates are denoted by CI_w , CI_f , CI_s and SR_w , SR_f , SR_s .

In Fig. 1(a), the CI scattering rates exhibit markedly different energy dependences in MLG and BLG. For BLG, the CI rate, CI_b , is relatively large at low carrier energy and decreases moderately with increasing energy. It is of the order of 10^{12} s^{-1} at low carrier energy and decreases to several 10^{11} s^{-1} at $E_k \simeq 0.35 \text{ eV}$. In contrast, the MLG CI rate, CI_m , exhibits a pronounced non-monotonic dependence. It increases rapidly at very low energy, reaches a maximum of approximately 10^{12} s^{-1} around $E_k \simeq 0.08\text{--}0.10 \text{ eV}$, and subsequently decreases with increasing carrier energy.

The different energy dependences of CI_m and CI_b arise from the distinct electronic structures of MLG and BLG. In MLG, the carrier wave vector, density of states, screening wave vector, and chiral overlap factor

System	Dispersion	$D(E_F)$	q_{TF}	$u(\theta)$	$u(\pi)$
MLG	$E_k = \hbar v_F k$	$\frac{g_s g_v k_F}{2\pi \hbar v_F}$	$\propto k_F$	$\frac{1 + \cos \theta}{2}$	0
BLG	$E_k = \frac{\hbar^2 k^2}{2m^*}$	$\frac{g_s g_v m^*}{2\pi \hbar^2}$	constant	$\frac{1 + \cos 2\theta}{2}$	1
2DEG	$E_k = \frac{\hbar^2 k^2}{2m_{2\text{DEG}}^*}$	$\frac{g_s m_{2\text{DEG}}^*}{2\pi \hbar^2}$	constant	1	1

Table 3: Comparison of the low-energy electronic properties relevant to charged-impurity scattering in mono-layer graphene (MLG), bilayer graphene (BLG), and a conventional two-dimensional electron gas (2DEG).

all contribute to the energy dependence of the scattering rate. In particular, $q_{\text{TF}}^{\text{MLG}} = 4r_s k_F$, so that the screening wave vector varies with k_F and hence with carrier energy. These combined dependencies produce the non-monotonic behavior of CI_m . In low-energy BLG, by contrast, the approximately parabolic dispersion gives an approximately constant DOS and a density-independent Thomas–Fermi screening wave vector. The resulting CI rate therefore exhibits a smoother energy dependence.

Around $E_k \simeq 0.1$ eV, the MLG and BLG CI rates become comparable in magnitude. At higher carrier energies, CI_b remains larger than CI_m . Thus, the relative importance of CI scattering in MLG and BLG is strongly energy dependent and reflects the combined effects of their different dispersions, DOS, screening properties, and chiral overlap factors.

The influence of dielectric screening on CI scattering is considered by comparing screened and unscreened relaxation rates. Screening reduces the strength of the long-range Coulomb interaction and consequently modifies the magnitude of the CI relaxation rate, particularly at small momentum transfer.[23, 32, 42] The comparison demonstrates that dielectric screening has a significant effect on the calculated CI rate and should therefore be retained when estimating realistic CI scattering in BLG rather than treating the Coulomb interaction as unscreened. The effect of dielectric screening on CI and SR scattering in BLG is shown in Fig. 1(b). The unscreened CI rate, CI_w , is substantially larger than the corresponding screened rates over the energy range considered. Inclusion of screening therefore significantly suppresses the strength of the long-range Coulomb interaction and also modifies the energy dependence of the CI relaxation rate.[57]

9.4 Acoustic Phonon Scattering

The AP transport relaxation rate in BLG was evaluated both with and without PB, using the analytical expressions derived in Eqs. (22) and (23), respectively. The corresponding numerical transport integrals were evaluated independently to assess the validity of the analytical results. The two approaches coincide within numerical accuracy over the investigated energy range. The calculated phonon-limited transport relaxation rates in BLG are summarized in Fig. 2(a), which shows the contributions from AP_b , OP_b , and SPP_b scattering as functions of carrier energy. The solid curves represent calculations without the PB factor, whereas the dashed curves include PB. The separation between the two curves therefore indicates the modification of the transport relaxation rate due to final-state occupation. The corresponding comparison between MLG and BLG is shown in Fig. 2(b), where the subscripts m and b denote MLG and BLG, respectively.

The calculated AP transport relaxation rate in BLG exhibits distinct absorption and emission contributions. The absorption contribution extends toward low carrier energies, whereas the finite-phonon-energy treatment imposes a kinematic threshold on phonon emission, since the carrier must possess sufficient energy to satisfy the emission condition. Consequently, the available phase space for emission is restricted near the threshold, giving rise to the pronounced low-energy structure observed in the AP contribution in Fig. 2(a). This behavior reflects the role of the phonon phase space and finite phonon energy in determining the energy dependence of AP transport relaxation.

The effect of PB is most pronounced in the low-energy region, where the solid and dashed AP curves in Fig. 2(a) show a noticeable separation. At higher carrier energies, the two curves remain relatively close, indicating that PB has only a modest effect on the AP transport relaxation rate over most of the investigated range. The inset of Fig. 2(a) provides a direct comparison of the AP transport relaxation rates calculated with and without PB and confirms this trend. The enhanced low-energy difference reflects the stronger

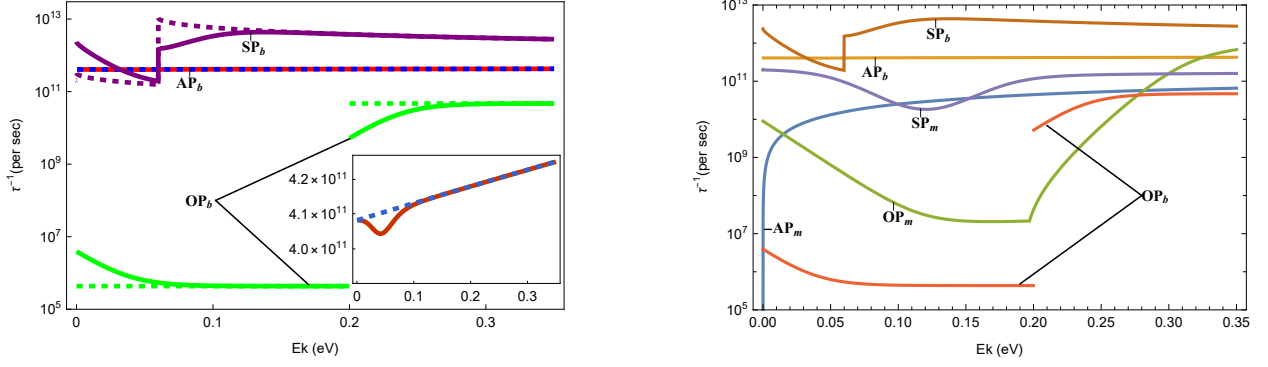


Figure 2: (a) Calculated carrier transport scattering rates, τ^{-1} , as a function of carrier energy E_k in bilayer graphene for AP (AP_b), optical-phonon (OP_b), and surface-polar-phonon (SPP_b) scattering mechanisms. Solid curves represent the results without Pauli-blocking (PB), whereas dashed curves include the PB factor. The inset shows an enlarged view of the acoustic-phonon contribution, highlighting its energy dependence in the range of approximately 0–0.35 eV. (b) Calculated carrier transport scattering rates, τ^{-1} , as a function of carrier energy E_k for monolayer and bilayer graphene due to acoustic-phonon (AP), optical-phonon (OP), and surface-polar-phonon (SPP) scattering mechanisms. The subscripts m and b denote monolayer and bilayer graphene, respectively. The figure compares the energy dependence and relative magnitudes of the phonon-limited scattering rates in the two graphene systems.

restriction of available final electronic states by Pauli exclusion. Thus, while PB does not substantially alter the AP transport relaxation rate over the full energy range, its inclusion is important for an accurate description of the low-energy scattering regime.

For comparison, the AP relaxation rate in BLG is considered alongside the corresponding behavior in MLG and a conventional 2DEG. The calculated MLG and BLG AP rates are shown by the AP_m and AP_b curves, respectively, in Fig. 2(b). In the high-temperature equipartition regime, where $\hbar\omega_{AP} \ll k_B T$, AP scattering may be treated in the quasi-elastic approximation.[36]

For MLG, the established Boltzmann-transport result is

$$\frac{1}{\tau_{AP}^{MLG}} = \frac{D_{AP}^2 k_B T}{4\rho v_l^2 \hbar^3 v_F^2} E_k, \quad (38)$$

which shows that $\frac{1}{\tau_{AP}^{MLG}} \propto E_k T$. This energy dependence follows from the linear Dirac dispersion $E_k = \hbar v_F k$ and the corresponding energy-dependent DOS. The monolayer chiral overlap factor $u_{MLG}(\theta)$ also enters the transport angular average. [36]

A more general treatment retaining the finite AP energy and PB was developed by Khatoon *et al.* [29]. Their semi-inelastic treatment contains separate phonon absorption and emission contributions and approaches the quasi-elastic result in the appropriate high-temperature limit.

For low-energy BLG, the corresponding quasi-elastic result obtained from the present formalism is

$$\frac{1}{\tau_{AP}^{BLG}} = \frac{D_{AP}^2 m^* k_B T}{4\rho v_l^2 \hbar^3}, \quad (39)$$

so that $\frac{1}{\tau_{AP}^{BLG}} \propto T$. Unlike MLG, the BLG rate is independent of carrier energy within the low-energy parabolic two-band approximation. This behavior follows from the approximately constant BLG DOS. [5, 32]

For a conventional parabolic 2DEG with no chiral overlap factor, the corresponding quasi-elastic result is

$$\frac{1}{\tau_{AP}^{2DEG}} = \frac{D_{AP}^2 m_b k_B T}{2\rho v_l^2 \hbar^3}, \quad (40)$$

and therefore $\frac{1}{\tau_{AP}^{2DEG}} \propto T$. [54]

Thus, the quasi-elastic AP relaxation rates exhibit the scaling

$$\frac{1}{\tau_{\text{AP}}^{\text{MLG}}} \propto E_k T, \quad \frac{1}{\tau_{\text{AP}}^{\text{BLG}}} \propto T, \quad \frac{1}{\tau_{\text{AP}}^{\text{2DEG}}} \propto T. \quad (41)$$

The identical temperature scaling of BLG and the conventional 2DEG arises from the equipartition phonon population, while their parabolic dispersions and approximately energy-independent DOS lead to an energy-independent relaxation rate within the quasi-elastic approximation. However, their absolute scattering rates differ because BLG possesses a nontrivial chiral overlap factor, $u_{\text{BLG}}(\theta_{kk'})$, whereas the conventional 2DEG has $u_{\text{2DEG}} = 1$. For the same deformation-potential coupling and other parameters, the BLG chiral factor reduces the transport angular average by a factor of two, leading to $\frac{1}{\tau_{\text{AP}}^{\text{BLG}}} = \frac{1}{2} \frac{1}{\tau_{\text{AP}}^{\text{2DEG}}}$.

The comparison therefore highlights three distinct regimes: MLG exhibits an energy- and temperature-dependent AP rate, whereas both BLG and a conventional 2DEG exhibit an energy-independent rate proportional to temperature within the quasi-elastic approximation. The difference between BLG and the conventional 2DEG then arises from the BLG pseudospin chirality, which modifies the transport angular weighting of AP scattering.

9.5 Optical Phonon Scattering

The BLG OP transport relaxation rate was evaluated both with and without PB, using the analytical expressions given in Eqs. (25) and (26), respectively. The analytical PB-inclusive result has been compared with the corresponding direct numerical evaluation and the two results coincide within the numerical accuracy over the investigated energy range.

The calculated BLG OP scattering rate, OP_b , is shown in Fig. 2(a), while the corresponding comparison between MLG and BLG is shown by the OP_m and OP_b curves in Fig. 2(b). A distinct change in the scattering rate occurs at approximately 3.77×10^{-13} erg, corresponding to the optical phonon energy used in the calculation. This feature separates the regions in which phonon absorption and emission contribute differently to the scattering rate. At very low carrier energies, approximately $0-2 \times 10^{-14}$ erg, the numerical scattering points are comparatively sparse, reflecting the restricted phase space associated with the finite OP energy. In particular, OP emission is kinematically allowed only when $E_k \geq \hbar\omega_o$, whereas absorption is allowed at all carrier energies in the idealized dispersion used here. The resulting energy dependence therefore differs qualitatively from that of AP scattering, for which the characteristic phonon energy is much smaller.

The influence of PB is examined in Fig. 2(a), where the OP relaxation rates calculated with and without PB are compared. In the Figure solid curves represent the results without PB, whereas dashed curves include the PB factor. The rate obtained without PB is comparatively insensitive to carrier energy except for the change associated with the phonon-emission threshold. Inclusion of PB introduces additional energy dependence through the occupation of the final electronic states. The difference between the two results is most apparent in the low-energy regime, where the electronic occupation factors have a stronger influence on the available final states. Thus, PB provides an important correction to the OP relaxation rate and should be retained when an accurate inelastic transport description is required.

The calculated OP relaxation rates of BLG and MLG are compared in Fig. 2(b). Under the parameters adopted in the present calculation, the BLG relaxation rate is lower than the corresponding MLG rate over the investigated energy range. This difference should not be interpreted solely as a difference in electron-phonon coupling strength, since the scattering rates also depend on the band dispersion, DOS, chiral overlap factor, and the specific electron-phonon coupling parameters used for each system.[1, 5] The comparison nevertheless demonstrates that OP scattering has a different energy dependence and magnitude in BLG and MLG.

This behavior should also be distinguished from the AP case. The relative magnitude of the BLG and MLG rates depends on the phonon branch and on the corresponding electron-phonon coupling matrix element; therefore, the ordering of the acoustic- and optical-phonon scattering rates need not be the same.

For comparison, consider first OP scattering in a conventional 2DEG with parabolic dispersion and an approximately energy-independent DOS. For a dispersionless OP and a momentum-independent electron-phonon matrix element, the transport relaxation rate including PB can be expressed as

$$\frac{1}{\tau_{\text{OP}}^{\text{2DEG}}} = \Gamma_o \left[N_o \frac{1 - f(E_k + \hbar\omega_o)}{1 - f(E_k)} + (N_o + 1) \frac{1 - f(E_k - \hbar\omega_o)}{1 - f(E_k)} \Theta(E_k - \hbar\omega_o) \right], \quad (42)$$

where Γ_o denotes the energy-independent scattering prefactor.

In the absence of PB, this expression reduces to $\frac{1}{\tau_{\text{OP}}^{\text{2DEG}}} = \Gamma_o [N_o + (N_o + 1)\Theta(E_k - \hbar\omega_o)]$. The two terms correspond to OP absorption and emission, respectively, with emission allowed only when $E_k \geq \hbar\omega_o$.

The conventional 2DEG and low-energy BLG both possess parabolic dispersions and approximately energy-independent DOS. Consequently, their OP scattering rates have similar phase-space characteristics. The principal distinction is that BLG contains the chiral overlap factor u_{BLG} , whereas no corresponding pseudospin factor occurs in a conventional 2DEG. [5, 54]

In contrast, MLG has the linear Dirac dispersion and an energy-dependent DOS. An analytical treatment of OP scattering in MLG including PB was given by Khatoun *et al.* [29]. Their result may be written, in the notation adopted here, as

$$\begin{aligned} \frac{1}{\tau_{\text{OP}}^{\text{MLG}}} = & \frac{V_o^2}{8\rho\omega_o\hbar^2v_F^2} \frac{e^{\frac{E_k}{k_B T}} \left(e^{\frac{E_F + E_k}{k_B T}} + 1 \right)}{e^{\frac{\hbar\omega_o}{k_B T}}} \\ & \times \left[\frac{(E_k + \hbar\omega_o)N_o}{e^{\frac{E_F}{k_B T}} + e^{\frac{E_k + \hbar\omega_o}{k_B T}}} \right. \\ & \left. + \frac{(E_k - \hbar\omega_o)(N_o + 1)\Theta(E_k - \hbar\omega_o)}{e^{\frac{E_F + \hbar\omega_o}{k_B T}} + e^{\frac{E_k}{k_B T}}} \right]. \end{aligned} \quad (43)$$

The first and second terms in Eq. (43) describe OP absorption and emission, respectively. As in BLG and the conventional 2DEG, emission is subject to the threshold $E_k \geq \hbar\omega_o$. Neglecting PB gives

$$\frac{1}{\tau_{\text{OP}}^{\text{MLG}}} = \frac{V_o^2}{8\rho\omega_o\hbar^2v_F^2} [(E_k + \hbar\omega_o)N_o + (E_k - \hbar\omega_o)(N_o + 1)\Theta(E_k - \hbar\omega_o)]. \quad (44)$$

The comparison among 2DEG, BLG, and MLG highlights the combined effects of band dispersion, DOS, and pseudospin chirality on OP scattering.[1, 5, 54] The 2DEG and low-energy BLG share parabolic dispersions and approximately constant DOS, whereas BLG additionally possesses the chiral overlap factor $u_{\text{BLG}}(\theta)$, MLG, in contrast, has a linear dispersion, an energy-dependent density of states $D(E) \propto E$, and the monolayer overlap factor $u_{\text{MLG}}(\theta_{kk'})$. Consequently, the OP relaxation rate has different carrier-energy dependences in the three systems. Nevertheless, the finite OP energy produces a common emission threshold, while PB modifies the available final-state phase space in all three systems.

9.6 Surface Polar Phonon Scattering

SPP scattering is an intrinsically inelastic carrier-relaxation mechanism, with distinct absorption and emission channels determined by the SPP energy $\hbar\omega_s$. [29] To examine the energy range over which these processes contribute to carrier relaxation, the BLG SPP scattering rate was evaluated both with and without PB using the analytical expressions given in Eqs. (28) and (30), respectively.

The BLG SPP scattering rate, SP_b , is shown in Fig. 2(a), while the corresponding MLG–BLG comparison is shown by the SP_m and SP_b curves in Fig. 2(b). The SPP rate exhibits a distinct change near the emission threshold. In the present calculation, this occurs at approximately $E_k \simeq \hbar\omega_s$, corresponding to the SPP energy used in the calculation. Below this threshold, only phonon absorption contributes, whereas above the threshold the phonon-emission channel becomes kinematically allowed in addition to absorption.

The opening of the emission channel produces a pronounced increase in the total SPP scattering rate. At higher carrier energies, the SPP rate remains significant, reflecting the continued contribution of inelastic phonon emission and absorption to carrier relaxation. The observed threshold behavior follows directly from the energy- conservation condition contained in the scattering integral.

In the present model, the distance-dependent factors e^{-2qb} and $e^{-2q(b+l)}$ account for the attenuation of the substrate-induced SPP electric field at the positions of the two graphene layers. For the parameter regime considered here, the layer-substrate separations are sufficiently small that $qb \ll 1$, $q(b+l) \ll 1$, and hence $e^{-2qb} \simeq 1$, $e^{-2q(b+l)} \simeq 1$. This approximation substantially simplifies the analytical treatment. The effect of retaining these exponential factors is nevertheless examined numerically to assess the validity of this approximation.

To further verify the analytical treatment, the fully numerical scattering integral was compared with the corresponding analytical expression. The overall agreement between the two results demonstrates that the analytical evaluation reproduces the numerical transport integral within the adopted approximations, with the largest differences occurring near the inelastic threshold.

SPP scattering in MLG has been studied using the Boltzmann transport formalism, including analytical treatments of the corresponding inelastic transport integrals in terms of complete elliptic integrals.[29] The present treatment extends this framework to BLG by incorporating the low-energy bilayer dispersion, DOS, and chiral overlap factor.

The mathematical structure of the BLG result remains closely related to that of MLG, with complete elliptic integrals appearing in the analytical evaluation of the angular dependence. Quantitative differences, however, arise from the distinct electronic structures. In particular, low-energy BLG possesses a parabolic dispersion and an approximately energy-independent DOS, whereas MLG has a linear Dirac dispersion and an energy-dependent DOS.

[1, 5] Furthermore, the BLG overlap factor $u_{kk'}$ differs from the MLG factor $u_{kk'}^{\text{MLG}}$. These differences modify both the magnitude and the carrier-energy dependence of the SPP transport relaxation rate.

The comparison also highlights the importance of PB. Because SPP scattering is inelastic, the initial and final electronic states generally have different energies, $E_{k'} = E_k \pm \hbar\omega_s$. Consequently, the occupation of the final electronic state can modify the scattering rate. The present BLG results show that the PB factor produces a measurable modification of the SPP relaxation rate, consistent with the role of final-state occupation in inelastic phonon scattering.

Thus, the present analytical treatment provides a BLG counterpart to the established MLG SPP framework,[29] while explicitly incorporating the parabolic low-energy BLG band structure and its associated chiral overlap factor. The resulting formulation can therefore be used to assess the contribution of substrate-induced polar phonons to carrier relaxation in supported BLG.

9.7 Comparative Analysis of Scattering Mechanisms

The preceding sections have developed analytical transport relaxation rates for the principal elastic and inelastic scattering mechanisms considered in BLG, including SR, AP, OP, SP, NI, and CI scattering. The resulting rates provide a basis for assessing the relative importance of these mechanisms under the parameter set adopted in the present calculations. The scattering mechanisms can be broadly divided into elastic and inelastic processes. NI, SR, and CI scattering are elastic within the approximations adopted here, whereas AP scattering is treated in the quasi-elastic regime and OP and SP scattering involve finite phonon energies and therefore inelastic carrier transitions. As summarized in Table 1, the relative importance of these mechanisms depends on carrier energy, temperature, carrier density, disorder strength, dielectric environment, and the relevant electron-phonon or impurity coupling parameters. Thus, the qualitative classification in the table does not by itself determine which mechanism dominates for a particular set of conditions.

Figure 3 summarizes the calculated transport scattering rates as functions of carrier energy and provides a direct comparison of the relative magnitudes and energy dependences of the six scattering mechanisms considered here. The figure shows distinct energy dependences for SRS, AP, SPP, NI, CI, and OP scattering over the investigated energy range.

A clear hierarchy among the different mechanisms is evident, although their relative importance is strongly energy dependent. NI scattering is among the dominant mechanisms over a substantial portion of the investigated energy range, particularly at intermediate and higher carrier energies. Its rate remains of the order of 10^{12} – 10^{13} s⁻¹ and exhibits only a moderate variation with carrier energy. This behavior is consistent with the approximately energy-independent low-energy DOS of BLG and the finite-range NI transport kernel discussed earlier. The approximately constant low-energy DOS associated with the parabolic BLG bands is a well-established feature of its low-energy electronic structure. [5, 58, 59]

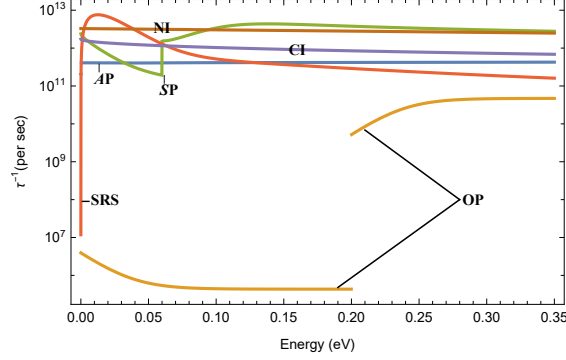


Figure 3: Calculated carrier transport scattering rates, τ^{-1} , as a function of carrier energy E_k in bilayer graphene for surface-roughness (SR), acoustic-phonon (AP), surface-polar-phonon (SPP), neutral-impurity (NI), charged-impurity (CI), and optical-phonon (OP) scattering mechanisms. The figure compares the relative magnitudes and energy dependences of the different scattering channels over the investigated energy range.

The AP and CI contributions are also substantial over the investigated energy range. The AP rate is comparatively weakly dependent on carrier energy within the quasi-elastic regime, consistent with the approximately constant low-energy DOS of BLG and the standard treatment of AP scattering in graphene-based systems. [36, 43, 60] The CI rate decreases progressively with increasing carrier energy in the present calculation, reflecting the combined effects of carrier wave vector, dielectric screening, and the chiral transport factor. CI scattering and its dependence on screening and carrier density have been extensively studied in MLG and BLG. [23, 51, 59] Thus, although NI scattering is one of the dominant contributions over a substantial portion of the energy range shown, CI and AP scattering provide significant parallel momentum-relaxation channels.

For the parameter set adopted here, the SPP contribution is particularly large in the low-energy region and exhibits a pronounced change near $E_k \simeq 0.06$ eV, associated with the finite SPP energy. As discussed in the SPP section, this behavior originates from the distinction between phonon absorption and the opening of the phonon emission channel. Above the emission threshold, the total SPP scattering rate is modified by the additional inelastic phase space. Consequently, the importance of SPP scattering depends not only on carrier energy but also on the dielectric environment and the SPP parameters of the substrate. [24–26, 48]

OP scattering is comparatively weak in the lower-energy part of the investigated range, primarily because the finite OP energy imposes a threshold for phonon emission. For an OP of energy $\hbar\omega_{OP}$, emission requires the initial carrier energy to satisfy approximately $E_k \geq \hbar\omega_{OP}$, subject to the detailed band structure and momentum-conservation conditions. Below this threshold, the emission channel is kinematically inaccessible, and the OP contribution is therefore strongly suppressed. Once the threshold is reached, the opening of the inelastic emission channel produces the rapid increase in the OP rate observed in Fig. 3. Thus, the low-energy weakness of OP scattering should not be interpreted as an intrinsically weak electron–optical-phonon coupling in BLG, but rather as a consequence of the finite phonon energy and the associated inelastic phase space.

The relative importance of AP scattering is further influenced by the low-energy electronic structure of BLG. Because BLG possesses an approximately parabolic low-energy dispersion and a nearly constant DOS, low-energy electronic states provide substantial phase space for AP scattering. In addition, the multilayer structure introduces additional acoustic-like phonon branches that can contribute to carrier scattering. First-principles calculations of electron–phonon interactions in BLG have shown that low-energy acoustic and acoustic-like phonon modes can make important contributions to carrier scattering, particularly at low and moderate carrier energies. [44] Consequently, the larger AP rate than OP rate observed below the OP threshold in the present calculation results from the combined effects of the quasi-elastic nature of AP scattering, the available low-energy phase space, and the finite threshold for OP emission. At sufficiently high carrier energy, however, OP scattering can become an important inelastic relaxation channel. [29, 44, 53]

SRS exhibits a strong energy dependence in the low-energy region. Its rate rises rapidly from very small values near zero carrier energy and then decreases progressively with increasing carrier energy. It therefore provides a non-negligible momentum-relaxation channel over a considerable portion of the investigated range. This behavior reflects the dependence of the SRS rate on the Fermi wave vector, the roughness correlation length, and the associated transport angular kernel. SRS and interface-induced potential fluctuations are recognized sources of momentum relaxation in supported low-dimensional systems, and their importance depends on the microscopic roughness parameters and device geometry. [22, 50, 61]

The results in Fig. 3 demonstrate that the dominant scattering mechanism is not determined by carrier energy alone but by the combined effects of electronic structure, disorder parameters, phonon populations, dielectric screening, and inelastic phase space. In particular, the present calculations show that SPP and SRS can make large contributions in the low-energy region, while NI becomes one of the dominant channels over a substantial intermediate- and higher-energy range. AP and CI provide significant competing momentum-relaxation channels, whereas OP scattering becomes important only after its corresponding inelastic emission threshold is accessible. The relative hierarchy therefore changes with carrier energy even for the fixed parameter set considered here.

10 Conclusion

In this paper, the carrier-scattering characteristics of bilayer graphene (BLG) have been investigated using complementary analytical and numerical approaches. The principal scattering mechanisms considered are neutral-impurity (NI), surface-roughness (SR), charged-impurity (CI), acoustic-phonon (AP), optical-phonon (OP), and surface-polar-phonon (SPP) scattering. Their energy-dependent transport relaxation rates were derived within the Boltzmann transport framework and evaluated numerically over a broad range of carrier energies. The analysis shows that the relative importance of these mechanisms is strongly dependent on carrier energy, temperature, screening, and phonon occupation, and therefore no single scattering mechanism universally dominates carrier relaxation in BLG.

The NI and SR mechanisms are found to be particularly important in the low-energy regime. NI scattering exhibits a characteristic energy dependence associated with the BLG density of states and chiral overlap, while SR scattering is strongly influenced by the finite correlation length of the graphene–dielectric interface and by dielectric screening. Screening substantially modifies the SR relaxation rate and must therefore be included when assessing interface-limited transport in BLG. The CI analysis further shows that charged-impurity scattering is strongly dependent on impurity density and screening, with its low-energy behaviour reflecting the electronic structure of BLG.

The phonon-mediated mechanisms exhibit distinct energy and temperature dependences. AP scattering is governed by the available phonon phase space and the energy dependence of the BLG density of states, while OP scattering is strongly constrained by the optical-phonon emission threshold and by Pauli blocking. Consequently, OP scattering becomes increasingly important once the carrier energy exceeds the relevant phonon energy. SPP scattering provides an additional inelastic relaxation channel in BLG on polar dielectric substrates. Its contribution is particularly important in the low-energy regime considered here and is strongly affected by substrate properties, phonon energy, carrier energy, and temperature. The comparison of the analytical and numerical SPP results also demonstrates the importance of retaining the full energy-dependent BLG density of states and chiral-overlap factors when evaluating the relaxation rate.

A comparative analysis with monolayer graphene (MLG) and a conventional two-dimensional electron gas (2DEG) further demonstrates that the scattering behaviour in BLG cannot be described solely by generic two-dimensional transport models. The approximately parabolic low-energy dispersion of BLG gives rise to an approximately constant density of states, while its chiral overlap factor introduces an additional characteristic dependence into the scattering rates. These features lead to energy dependences that differ from those of both MLG and a conventional parabolic 2DEG. The comparison therefore highlights the importance of incorporating the actual electronic structure of BLG when modelling carrier relaxation.

Overall, the results establish that carrier transport in BLG is governed by a competition among elastic and inelastic scattering mechanisms whose relative contributions vary with the physical and material parameters of the system. The analytical expressions and numerical results presented here provide a unified framework for identifying the dominant relaxation channels under different transport conditions and for assessing the

roles of disorder, dielectric screening, and phonon-mediated processes in BLG-based electronic and high-frequency devices.

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